

AI for Financial Practitioners

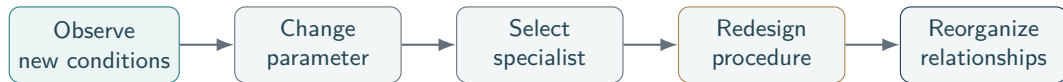
Lecture 3 · Part I

When the Organization Becomes a Variable

Learning, adaptation and reorganization without allowing improvement to rewrite institutional purpose

After delegation comes change

Once authority has been delegated, the next governance problem is deciding what the system itself may change.



The opportunity

Markets, data formats, workloads and failure patterns change. A fixed organization may become inefficient.

The danger

A system can appear to “improve” by changing the objective, acceptance test, permissions or definition of success.

Change itself must become an explicit governed object

Feedback is not adaptation

Feedback is evidence. Adaptation begins only when an authorized process uses that evidence to change a defined object.

Operating loop — today's work

- ▶ New filing or market event arrives
- ▶ Retrieve and reconcile evidence
- ▶ Calculate controlled measures
- ▶ Interpret and act within current authority
- ▶ Escalate exceptions



Review loop — tomorrow's process

- ▶ Accumulate failures and corrections
- ▶ Diagnose recurring mechanisms
- ▶ Propose a bounded modification
- ▶ Test against baseline and protected cases
- ▶ Release only through authorized approval

A single exception calls for escalation. A pattern of exceptions may justify redesign.

What may change — and what must be protected

Adaptive systems need an explicit boundary between mutable means and protected institutional commitments.

MUTABLE COMPONENTS

- Routing parameters
- Specialist selection
- Prompts and extraction procedures
- Memory procedures
- Workflow sequence
- Candidate models

PROTECTED REQUIREMENTS

- Institutional objective
- Acceptance criteria / evaluator
- Data permissions
- Action permissions
- Risk and resource limits
- Release authority

The system may search for better behavior. It may not silently redefine what “better” means.

The objects of learning

Governance starts by naming exactly what changes, because different change objects require different evidence and authority.

Object	Example in finance	Governance question
State / facts	New issuer data; new prices	Is behavior changing under a fixed policy?
Parameter	Alert threshold; routing score	What range is pre-authorized?
Procedure	Retrieval or reconciliation skill	Who validates the revised process?
Organization	Specialist / agent relationships	What dependencies and concentration emerge?
Evaluator	Test set; acceptance metric	Is evaluation independent of the candidate?
Objective / authority	Risk limit; action scope	Requires separate institutional decision

Do not call all behavioral change “learning”

Recursive improvement: improving the improver

When the improvement process itself becomes mutable, the evaluator and governing objective must remain independently protected.



Recursive layer

The system may compare alternative ways of generating improvements: critics, clustering failures, adversarial tests, new proposal workflows.

Protected envelope

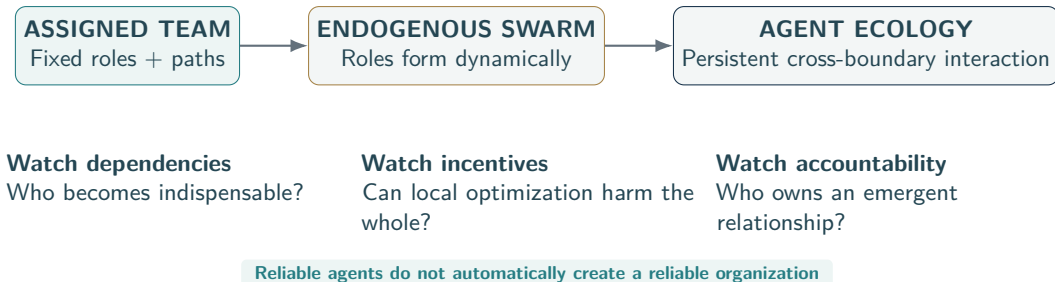
Candidate changes cannot rewrite the held-out tests, constraints, purpose or release authority used to judge them.

Recursive improvement is an architecture to

test — not an assumption of inevitable progress.

From teams to swarms and ecologies

When roles and relationships become mutable, the organization itself becomes part of the model risk.



Test the mechanism, not the story

A credible adaptation experiment states the causal mechanism, comparison, acceptance criteria and result that would disconfirm it.

FIXED BASELINE

Same specialist structure

ADAPTIVE

Activate specialist selectively

ALWAYS ACTIVE

Specialist on every case

Measure more than quality

Held-out cases · difficult regimes · latency · cost · failure frequency · reviewer burden · ability not to intervene.

Ask the causal question

Did adaptation create value, or did simply adding more capacity create the apparent improvement?

A

result with no possible disconfirming outcome is a story, not an experiment.

Worked case: adaptive credit triage

Adaptive routing becomes institutionally credible when it is bounded, benchmarked, observable and reversible.

Hypothesis

Dynamic routing uses event text, issuer state and prior corrections to reduce reassignment and completion time.

Protected

Approved sources · calculation definitions · escalation thresholds · specialist tools · human portfolio authority.

Evaluation

Historical replay → shadow mode → bounded release. Measure reassignment, useful-completion time, evidence defects, corrections, cost and workload concentration.

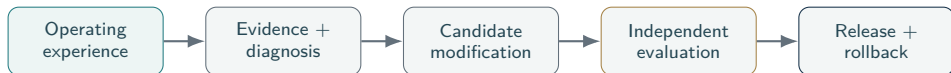
Rollback

Repeated overrides, category drift, specialist overload or protected-case failures restore the fixed router.

The experiment tests a mechanism — not “self-organization” as a narrative

The organization becomes a governed variable

The institution can permit rapid search for better behavior while reserving purpose, evaluation and authority for accountable governance.



Let the system search for better behavior inside a boundary whose purpose it cannot silently rewrite.

Leadership view

Maintain a change register: change object, protected requirements, owner, evaluator, evidence, stage and rollback status.

Bridge to Part II

The next problem is institutional: how can the **Fortress** protect commitments while the **AI Factory** experiments and creates the next capability?